Bank Seed 20261026 — Validation Note

Date · 2026-10-24

File · `bank-seed-20261026.json` — 909,289 bytes

SHA-256 · `e16e1080b55514e72af520c7b8f2ab0faf2b9491fd3d7b81b8acdd3de8b14030`

Produced by · Muse, running Charlie's engine (`fs_quiz`, from the `ncees-exam-prep` monorepo, `engine/src`) — `generate --all --n 25 --seed 20261026`

Replaces · `bank-seed-20261021.json` (480 questions) in the app at the 0.9.2 swap

Result

1,143 questions, schema v1, engine validator: VALID.

Raw generation produced 1,200 (48 topics × 25). 57 questions were exact duplicates (identical stem + choices + answer — an artifact of re-drawn conceptual templates) and were removed; the file's `question_count` was corrected to match. Every remaining question is a distinct stem/choice/answer triple.

Domain · Before (480) · After (1,143)

survey-computations · 120 · 300

boundary-law-and-real-property · 70 · 174

surveying-processes-and-methods · 70 · 164

applied-math-statistics · 60 · 148

mapping-processes-and-methods · 60 · 134

surveying-principles-geodesy · 50 · 116

business-concepts-ethics · 50 · 107

Difficulty mix: 414 / 679 / 50 (levels 1 / 2 / 3) — same profile shape as the current bank.

Checks run

- Engine's own contract validator (`fs_quiz validate`): **VALID — schema v1, 1143 questions, 7 domains**
- All qids unique; every question has exactly 4 choices, a valid `answer_index`, and a non-empty explanation
- Per-question keys byte-identical in structure to the current bank's (the app's loader contract is unchanged)
- Zero domains regressed vs the 480 bank

Honest caveats (for the record)

- **Conceptual topics have small unique pools.** Business ethics, datums, lidar-type generators vary distractors more than substance; some questions repeat a stem with re-rolled choices. That matches the current bank's behavior (its 50 business questions rest on ~14 unique stems) — variants are legitimate practice reps, but growing *unique* conceptual content past this point requires new generator templates, i.e., engine development, not bigger draws.
- **Generator fragility found:** at N=25, 3 of 4 attempted seeds crashed in a geodesy generator (`ValueError: need 4 unique choices` — distractor pool collapse). Seed 20261026 completed cleanly. Logged here as an engine-backlog item; it does not affect this bank, which the validator passed end-to-end.
- 68 stems also appear in the old bank (same templates, different numbers by seed) — expected and harmless.